VERSION
14.0



Resource Management

AFLCMC/WA

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AMS GRA ARCHITECTURE VOLUMES

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1.0 Overview

Resource Management is the process of allocating, scheduling, and controlling execution as well as managing the resources of the Agile Mission Suite (AMS) Government Reference Architecture (GRA) in order to maximize mission effectiveness. The AMS GRA includes Sensor/Effector Resources, Computing Resources, and Mission System Resources.

1.1 Purpose

The Resource Management Volume describes how the AMS GRA can implement software in order to effectively manage the resources of the Mission System including Sensing and Processing elements.

1.2 Scope

The Resource Management Volume aids the Mission System Integrator (MSI) in designing the specific set of services used to implement Resource Management.

1.3 Informative References

Informative references are supplemental documents that a procuring authority, MSI, or Minimum Procurable Unit (MPU) provider refers to for more information about an AMS GRA entity or related entities. See **Table 1.3.1** for a list of informative references for the Mission System Volume.

Standard	Title	Version/Date
AMS GRA Volumes	MFA MPU Volume	14.0
AMS GRA Volumes	AMS GRA Service MPU Volume	14.0
AMS GRA Volumes	Software Architecture Volume	14.0
AMS GRA Volumes	Mission System Volume	14.0
AMS GRA Volumes	OMS Service MPU Volume	14.0
AMS GRA Volumes	MASI Volume	14.0
AMS GRA Volumes	IR MEL IDD	14.0

Table 1.3.1: Informative References Identify Documents a Procuring Authority, MSI, or MPU Provider Can Use for More Information about Resource Management and Related Entitites.

2.0 Resource Management

The AMS GRA enables management of Mission System resources. A modern Mission System entails a broad scope of resources to be managed. These resources range from an MFA timeline to signal processing capacity.

Mission System resources generally fall into one of four categories:

1. *Resources*, such as Multi-Function Apertures (MFAs) and their constituent parts, Digital Payload Processors (DPPs), network switches/ports, memory, data storage, and power consumption
2. *Capacity Measures* on hardware resources, such as timeline, bandwidth, amount of memory, power, and cooling
3. *Skills and Services* that consume hardware resources
4. *Data* such as the products of MFAs or Skills

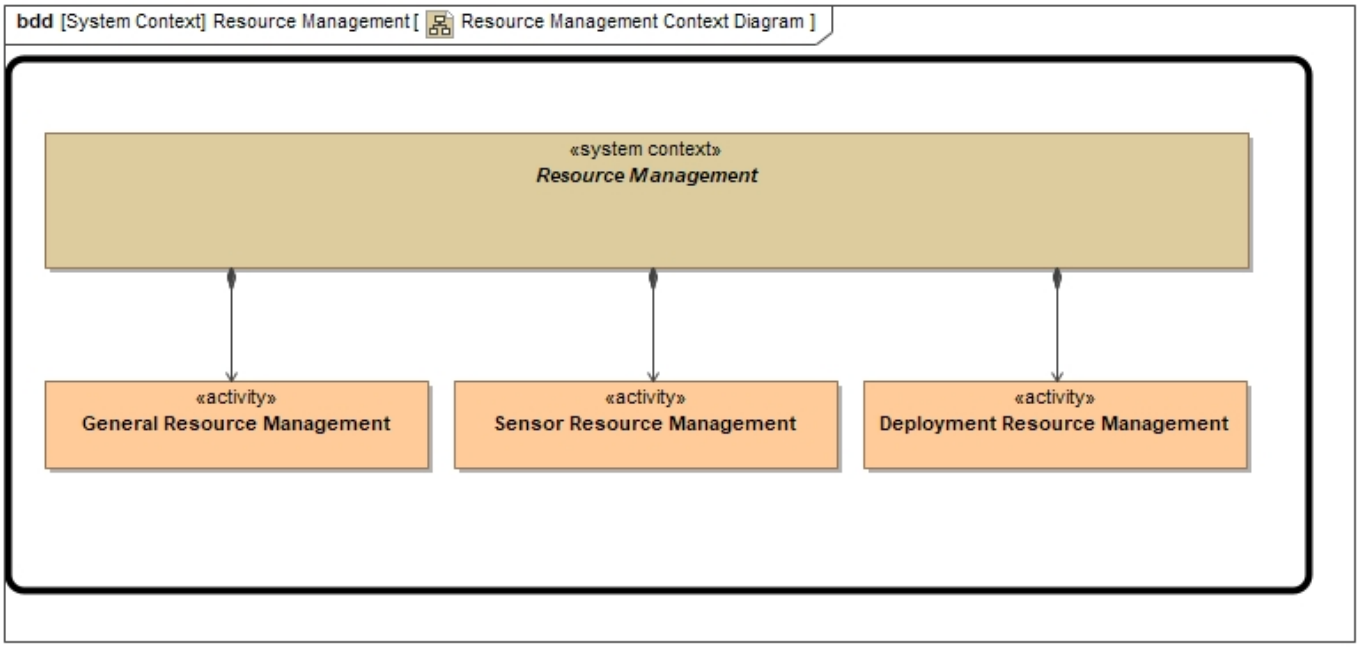


Figure 2.0.1: AMS GRA Resource Management Is Organized Into General Resource Management, Sensor Resource Management, and Deployment Resource Management Categories.

Deployment Resource Management

Deployment Resource Management comprises initializing Skills, provisioning these Skills with necessary processing and memory allocations, establishing the data paths required to operate, and evaluating the overall Mission System processing capacity versus information needs. The details of Deployment Resource Management can be found in **Section 2.3**.

Sensor Resource Management (SRM)

SRM provides interfaces and resource management functionality to enable effective runtime sharing of MFAs with multiple Skills. SRM optimizes operation to meet the information needs of the Skills or a data request from a higher layer (see **Section 2.1.1**). SRM generally interacts with mission-specific Skills (refer to the Software Architecture Volume for more details). The multi-layer SRM span-of-control includes the following:

- Manages compute resource requests and hardware for deployed Skills
- Deconflicts Mission Specific Capability Priorities
- Manages Spatial Requirements
- Manages Spectral Requirements
- Manages Thermal Limits
- Manages COSITE, Inhibit, and Blanking
- Manages Mission System Power
- Manages Aperture Resource Allocations

General Resource Management

General Resource Management addresses multi-ship and own-ship Resource Management. Considerations of negotiating conflicting resources between Security and mission critical needs may also fall into this category when within the jurisdiction of the Mission System. The General Resource Management span-of-control includes the following:

- Manages Multi-Ship Resource Management
- Manages Own-Ship Resource Management
- Balances Security and Mission Critical Needs
- Manages Prime Power Utilization
- Manages Multi-Ship Content Routing
- Coordinates Multi-Ship Sensing
- Manages Inter-Ship Communications Capacity

2.1 General Considerations

2.1.1 AMS GRA Functional Layering (Non-Prescriptive)

The AMS GRA Loosely-Coupled Layered Resource Management topology, illustrated in **Figure 2.1.1**, provides the isolation needed among key architectural elements to enable rapid development, integration, and fielding of Skills.

Key aspects of this architectural approach include:

- Each layer comprises a logical scope of responsibilities.
- Each element within a layer has a span-of-control over its internal elements and the related elements in the hierarchy.
- Each element within a given layer provides an opportunity for rapid technology insertion.
- Elements have diverse implementations tailored to the mission needs they address.

Loose coupling is essential for elements across a given layer in order to facilitate both timely decision-making during mission execution and rapid upgradeability between missions. Higher-level elements focus on conflict avoidance and prioritization of tasking among lower-level elements without constraining how those elements perform conflict resolution and optimization within their own spans-of-control. This hierarchal approach reduces coupling between the layers, speeds local decision making by providing the freedom to make the decision close to the problem, and enables improvement in performance through technology insertion into the specialized layers without disrupting other elements.

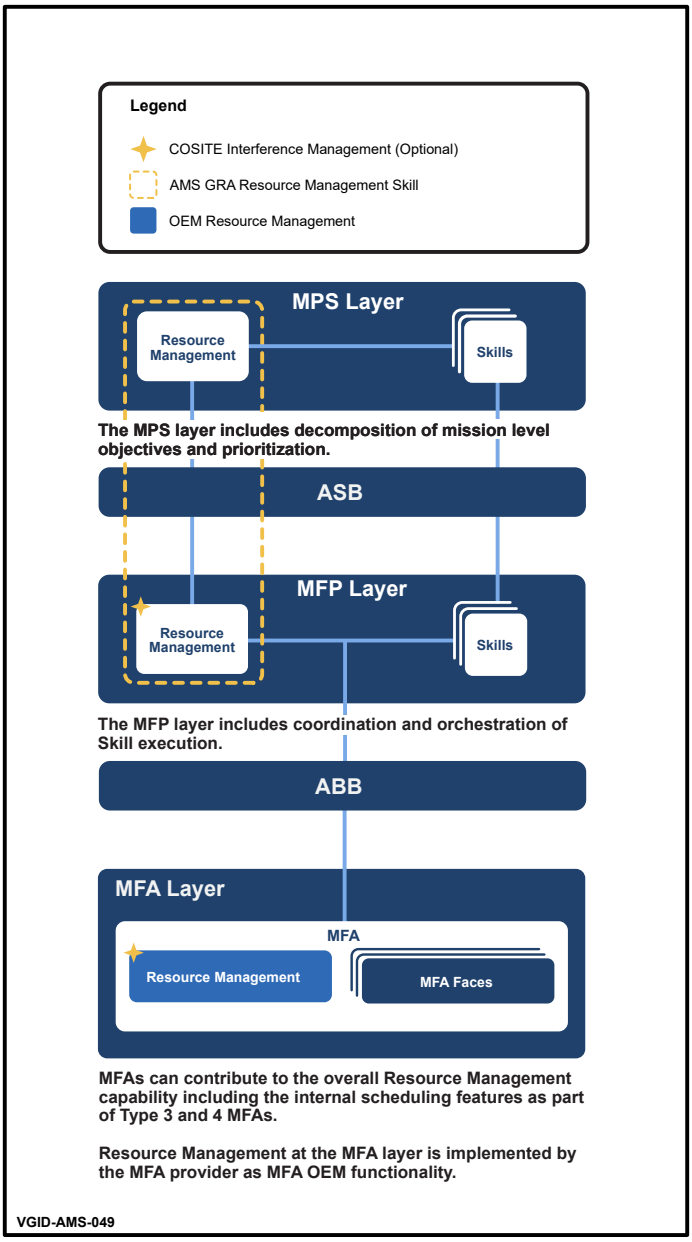


Figure 2.1.1: Resource Management Responsibilities, When Embodied in Loosely-Coupled Layers with Well-Defined Scope, Further the AMS GRA Mission for the Rapid Development, Integration, and Fielding of Skills.

Implementations may deviate from the strict hierarchical structure when performance improvement—achieved by sharing resources in a lower layer—justifies increased complexity. When these deviations occur, potential conflicts must be avoided and/or resolved in at least one of three ways:

- The first approach to avoid or resolve a conflict is via direction of a higher-layer element. For example, if two Multi-Function Processor (MFP) layer elements access the same MFA, the Mission Processing Subsystem (MPS) layer may task the MFP layer elements such that they do not have conflicts.
- The second approach manages larger scale resource sharing between Skills. This approach to resolve conflict ensures MFA dwell allocations are commensurate with Skill priority and MFA availability.
- The third approach provides lower layer elements of the architecture with enough information to resolve the conflict.

The interfaces between the layers use, at a minimum, Open interfaces consistent with AMS GRA architectural elements. In some cases, AMS GRA defines these interfaces as Common. For example, the AMS GRA has defined Common Radio Frequency (RF) and Infrared (IR) MFA Encapsulation Layer (MEL) interfaces, which provide abstractions for Skill and MFA interactions. This layered approach enables compliance to the Common interfaces, without driving large numbers of optional fields back into the MEL interfaces, such as messages that would be required for communication with the MPS layer.

The MFP and MFA layers seek to minimize interfaces between independently procurable elements. For example, Skills can be procured, deployed, and tasked independent of other Skills. MFAs should not be dependent on the implementation, state, or current behavior of other MFAs.

The partitioning and allocation of functionality and span of control associated with the Resource Management layers in AMS GRA is informed by the guiding attributes in **Table 2.1.1**, **Table 2.1.2**, and **Table 2.1.3**. These guiding attributes include time constraints, level of data processing and aggregation, and the span-of-control of elements within the layer.

Notional Time Constraint	Data Processing/Aggregation	Span-of-Control
Engagement to Mission Phases	Sensor Data Products to System Level Tracks	Entire Mission System Suite Multi-source integration and platform priority setting Multi-ship Prime power utilization

Table 2.1.1: The MPS Layer's Logical Scope of Responsibilities Is Derived by the Layer's Guiding Attributes.

Notional Time Constraint	Data Processing/Aggregation	Span-of-Control
Dwell or Timeslot to Multiple Scans	Conditioned to mission-specific Skill logic and processing of sensor data including raw and conditioned data products (e.g. Measurement reports and Sensor Tracks)	Logical groupings of Skills/services One or more MFAs per grouping Response modulation to near real-time data processing

Table 2.1.2: The MFP Layer's Logical Scope of Responsibilities Is Derived by the Layer's Guiding Attributes.

Notional Time Constraint	Data Processing/Aggregation	Span-of-Control
Intra-pulse to Dwell or Timeslot	Raw digitized data to conditioned data (e.g. beamformed, decimated, channelized, other data products via Local Function)	One or more aperture Faces Conditional pre-approval Low latency actions Front-end timeline management

Table 2.1.3: The MFA Layer's Logical Scope of Responsibilities Is Derived by the Layer's Guiding Attributes.

2.1.1.1 Mission Processing Subsystem Layer

The MPS layer's span-of-control covers the entire Mission System. The MPS generates platform-level information needs, tasking, and constraints. The MPS layer uses current mission state and objectives to determine prioritization provided to MFP layer Resource Management and Skills. This guidance flows to MFP and MFA layers and shapes overall Skill activity and mission execution timelines.

2.1.1.2 Multi-Function Processor Layer

The MFP layer Resource Management has a logical span-of-control of the Mission Specific Sensor Skills that utilize one or more MFAs. The functions associated with MFP layer Resource Management are performed by (potentially) multiple MPUs within the MFP layer dependent on requirement allocation and flow down from the MSI. For example, an individual Sensing Skill, using the RF MEL interfaces, may make decisions regarding Virtual Aperture (VA) selection based on specific performance metrics, flow down from the MSI, and constraints included with a tasking request. In parallel, a centralized RF compatibility function may manage transmit inhibits across a suite of MFAs. The compatibility function use the inhibit VA interfaces in the RF MEL. MFP layer Resource Management may also take advantage of the Universal Command and Control Interface (UCI) messages for RF Resource Allocation to provide a cross Skill approach which influences or directs VA selection for individual Sensing Skills. These Resource Management functions collectively provide mechanisms to facilitate the coordination of the set of deployed Sensing Skills and the shared usage of MFA resources.

Skills at the MFP layer are responsible for adhering to the priorities, constraints, and execution guidance (e.g. timing, spectrum usage, etc.) included in tasking from the MPS layer. These parameters are specified in the UCI messages. These messages are used for tasking Sensing Skills. Interfaces in the MEL provide a mechanism for Sensing Skills to communicate a subset of these parameters to the MFA, including settings for priorities, timing, and spectrum usage. AMS GRA establishes interfaces for Skills within the MFP layer to accomplish commands as received from the MPS. The AMS GRA does not employ methods to estimate the benefit of executing alternative activities if the original request cannot be fulfilled.

The MFP layer Resource Management has the additional responsibility to provide scheduling for Type 2 and Type 5 MFAs – which are defined as MFAs with no internal scheduler and one or more faces.

2.1.1.3 Multi-Function Aperture Layer

The MFA layer Resource Management manages internal MFA resources such as individual Faces and RF elements. The MFA abstracts group these resources into VAs and presents the VAs to the MFP layer through the MEL. This approach allows the MFA to implement MFA functions, while still allowing MFP layer Resource Management to manage Skills' use of MFA resources at the VA level of abstraction. MFA functions include local short-term scheduling, local arbitration, and translation of abstract actions to low-level/high-speed controls with cognizance of MFA specifics as necessary. Examples of MFA specifics include power and cooling dynamics, and loads and manages VA and Local Function definitions requested by Skills. VA definitions are defined In Accordance With (IAW) the MFA specific VA Database and VA Development Tool.

The MFA layer scheduling operates at the lowest scheduling granularity in the system. It fine-tunes the MFP layer schedule of MFA utilization right before the array executes service actions. MFA layer scheduling applies to Type 3 and Type 4 MFAs – which are defined as MFAs with internal scheduling and one or more faces.

For more information about MFA types, roles, and functions, refer to the key features defined in the MFA MPU Volume.

2.1.2 COSITE and Other RF Mitigation (Non-Prescriptive)

RF interoperability is essential to effective operation of the distributed MFAs in an AMS GRA system, as well as non-AMS GRA RF equipment on the platform. An interoperability scheme is necessary to manage interactions where one action disturbs another. These interactions may occur on the same MFAs, among multiple MFAs, or among distributed RF equipment on the platform. General interoperability schemes may span across the AMS GRA layers.

AMS GRA uses the Open Mission Systems (OMS) Critical Abstraction Layer and Common RF MEL to control and coordinate the behavior of Sensor Skills using shared MFA resources. These interfaces provide the means for the MPS to prioritize, constrain, and direct the activities of Sensor Services in the MFP layer. Comparably, the Common RF MEL provides interfaces for the MFP layer services to ensure that tasking direction and constraints continue to be communicated down to the MFA.

For RF compatibility, the UCI messages enable the MPS to direct Skills to use specific spectral ranges, control timing, and communicate priorities. These high-level parameters can be broadly applied at the Capability level or for specific commands. RF Resource Allocations can influence VA selection either within an MFA or across MFAs. The Common RF MEL defines interfaces that support functions such as transmit inhibit and RF blanking. These interfaces enable information about RF activities on one MFA to be communicated to multiple services

that may be impacted by transmissions. These interfaces also offer a mechanism for one or more services to inhibit transmit functions on specific MFAs.

AMS GRA does not prescribe the detailed behaviors and functional allocation for RF compatibility. Instead, AMS GRA enables multiple implementation patterns, ranging from a centralized paradigm to a broadly distributed operation. The allocation of specific functions is driven by operational mission, system latencies, required response timing, and details of MFAs and services within the Mission System.

Legacy and external equipment may impact, or be impacted by, the RF interactions of the AMS GRA Sensor System. COSITE mitigation techniques may include a priori or low latency mechanisms from external equipment.

There are opportunities for COSITE mitigation within each AMS GRA layer. The MPS layer may provide Capability Commands that balance RF usage where spectral content overlaps. The MFP layer may manage RF allocations of duty-cycle, blanking percentage, and opportunities of scheduling disruptive events. At the MFA layer, Resource Management may coordinate VA usage, schedule activities in a manner that prevents hardware damage, enable low-latency actions such as blanking and transmit inhibit, and interrupt in-progress RF activities with high priority, fast-response functions.

2.2 Sensor Resource Management (SRM)

SRM addresses the run-time operation and optimization of Skills using MFAs, as well as managing RF spectrum across both AMS GRA apertures and other platform emissions. AMS GRA defines a layered, hierarchical SRM topology structure that enables rapid insertion of diverse capabilities and technologies. This distributed approach for SRM is not intended to flow all decisions up to the highest layer.

2.2.1 Structural Architecture

The SRM Structural Architecture presented in **Figure 2.2.1** highlights the interfaces to other AMS GRA elements. A description of each element is listed in **Table 2.2.1**. Interactions are captured behaviorally in **Section 2.2.2**, and messaging associated with these interfaces is listed in **Section 2.2.3**.

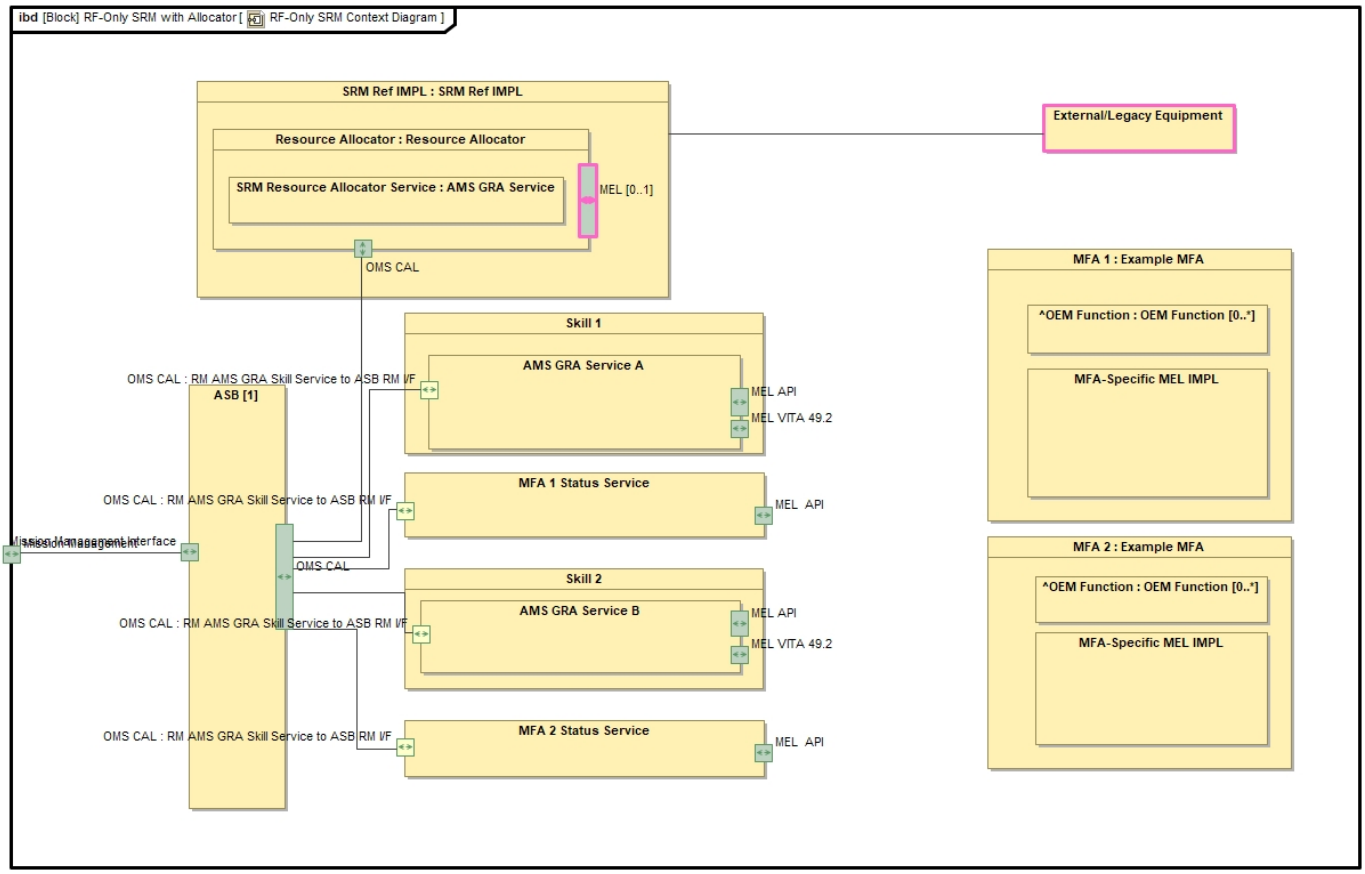


Figure 2.2.1: This SRM Structural Architecture Diagram Shows an Example Reference Implementation of RF-Only SRM.

Name	Documentation
SRM Resource Allocator Service	A software service within the Resource Allocator providing the business logic for processing resource requests and performance metrics, as well as awarding resource allocations on the basis of mission priorities, Skill capability needs, and Skill quality-of-service metrics.
SRM Ref IMPL	An aggregation of OMS and AMS GRA Services that collectively performs the function of an SRM reference implementation.

Name	Documentation
Skill 1	A reference collection of OMS and AMS GRA services, composing a Mission Specific Sensor Skill and utilizing an MFA. It requests MFA resource allocations from the SRM Resource Allocator Skill to fulfill Capability Commands.
Resource Allocator	Accepts Skill resource requests and allocates resources to Skills based on mission priorities and resource availability.
OEM Function	The OEM Function is software or firmware that is specific to an MFA and developed by the OEM. It does not support any third-party replacement.
MFA-Specific MEL IMPL	The software implementation of the Common RF MEL interface, provided by the MFA developer; it includes the MEL API library. This implementation realizes both the C2 and data elements of the RF MEL. It translates from the Common RF MEL definition to MFA specific OEM infrastructure.
MFA 1 Status Service	An AMS GRA Service that translates MEL MFA Health and Status messages to OMS Critical Abstraction Layer Subsystem Status messages and translates OMS Critical Abstraction Layer state control messages to MEL messages.
MFA 1	A reference implementation of an MFA. *See Multi-Function Aperture MPU Volume*.
External/Legacy Equipment	By discrete and/or messaging interface, external and legacy equipment on the platform provides interface to SRM indicating blanking or Tx inhibit events for spectrum deconfliction, or other on-board COSITE control. Examples include MIDS-JTRS, Legacy IFF, and navigation equipment.
ASB	The ASB is the bus supporting data exchanges between the Resource Management in the MPS layer and in the MFP layer.
AMS GRA Service A	A reference implementation of an AMS GRA Service.

Table 2.2.1: This Table Lists SRM Elements and Details Each Element's Role in RF-Only Resource Management.

The SRM Framework uses the existing OMS Critical Abstraction Layer interface between Skills and Mission Management services. For a given instantiation, this interface is intended to use a Common set of messages. Evolution of SRM reasoning is enabled by structuring the components similar to a Skill. Personality-based SRM services may be deployed according to the combinations of active Skills.

The primary function of SRM is to allocate resources through a decoupled and modular mechanism of load balancing, time sharing, and spatially separating conflicting jobs across time-sensitive, performance-demanding Skill activities. SRM control pathways may be distributed hierarchically between MPS, MFP, and MFA layers to improve responsiveness.

The Resource Allocator primarily communicates over the Abstract Service Bus (ASB) using UCI messages. It interacts with Sensing Skills using Resource Allocation messages to allocate MFA resources at timelines aligned with Capability Commands. If the Resource Allocator interfaces with an MFA, it does so using the MEL.

VA Job Requests from the Skills are directly routed to the MFA. A service uses the MEL library, delivered with a specific MFA, to interact with the MFA. For example, VA Job Requests from the Skills are directly routed to the MFAs Original Equipment Manufacturer (OEM) functions by the specific MEL Implementation (IMPL). The MEL IMPL presents the common RF MEL Application Programming Interface (API) to the Service then converts MEL API calls into MFA OEM protocol. For direct MFA connections the MEL IMPL is exercised to convert MEL API into OEM protocol.

AMS GRA comprises mission-agnostic and mission-specific Skills (see the Software Architecture Volume) that combine to provide the overall functionality required for an AMS GRA implementation. Resource Management algorithms may exist in both of these broad categories of software. Allocation of Resource Management functionality to specific services considers multiple factors. These factors include an assessment of whether the functionality is specific to a defined mission set or aligns to capabilities that are mission-agnostic. This approach enables software reuse of a core set of services.

As a general pattern, Resource Management functions and interfaces that are applicable across AMS GRA Mission Systems and missions belong in mission-agnostic Skills. A mission-specific Resource Management Skill likely provides Resource Management functions tied to specific missions or mission Skills. AMS GRA Compute Environment software provides the Resource Management functions coupled to specific hardware as part of implementation. Multiple pieces of AMS GRA Compute Environment software and Skills may implement the same Resource Management functions and interfaces.

2.2.2 Behavioral Architecture

The interactions of principal SRM Actors and Components in **Table 2.2.1** primarily follow an informed delegation pattern. In this pattern, once firm resource allocations have been coordinated and granted by the Resource Allocator, then Skills interact with the MFA directly.

The Resource Management Framework defines a basic set of interactions and information exchanges between elements of the SRM Structural Architecture, including UCI message exchanges between the Sensor Skills and the Resource Allocator.

Messages depicted in the sequence diagrams are described in **Table 2.2.2**. For simplicity, the diagrams depict interactions of a single Skill. The normative interaction patterns also apply with multiple Skills participating concurrently. As multiple Skills compete for limited resources, required resource allocation adjustments or revocations are determined by Resource Management and communicated to Skills as necessary.

The remainder of **Section 2.2.2** contains a non-exhaustive set of sequence diagrams which illustrate common SRM behaviors.

2.2.2.1 Mission Task Instantiation/Execution

Figure 2.2.2 shows the sequence diagram for instantiating a single mission task.

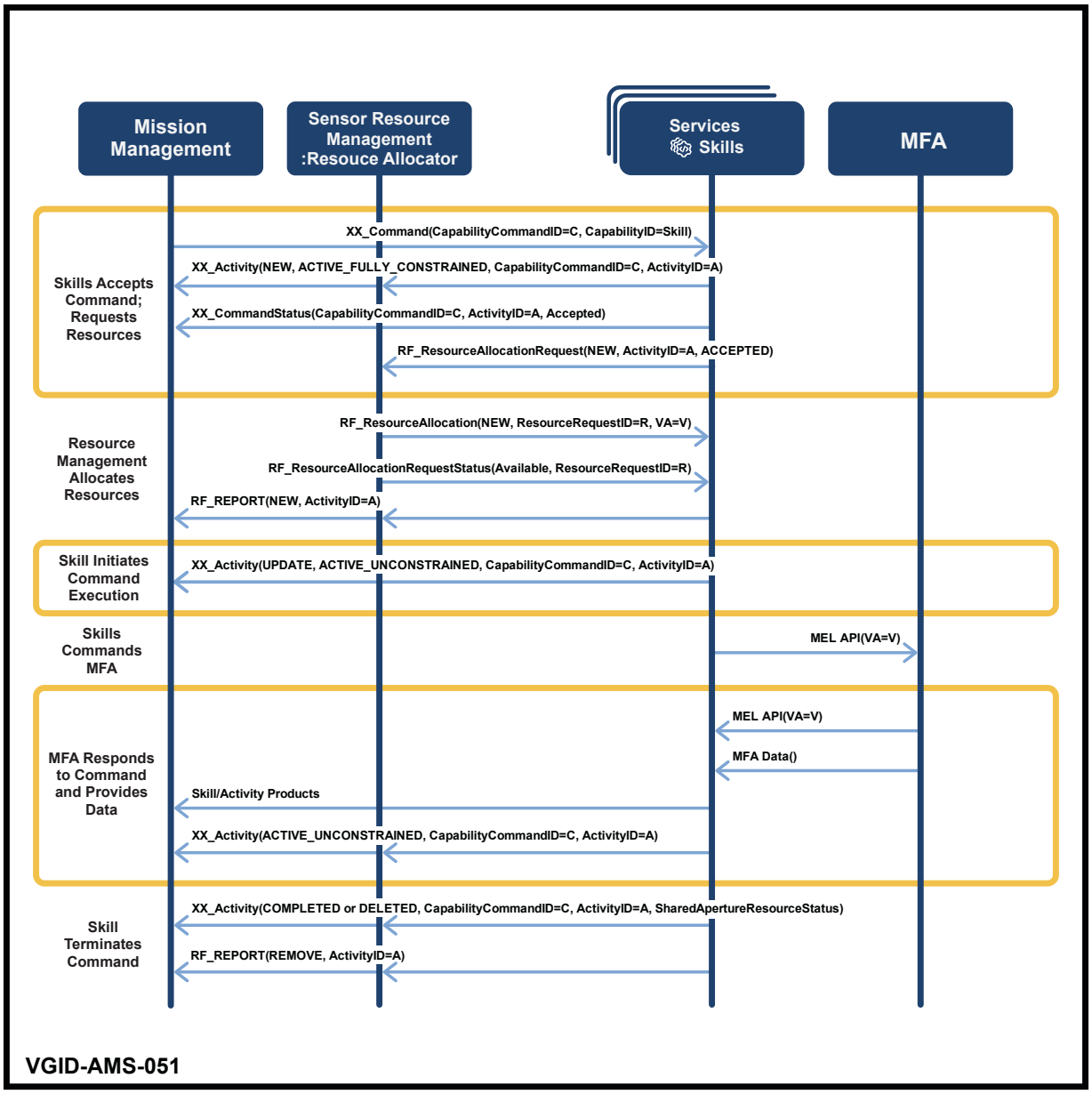


Figure 2.2.2: This SRM Sequence Diagram Shows The Lifecycle of a Mission Task.

This sequence provides an example of the information exchanges between architecture elements defined in the SRM Structural Architecture. However, the sequence does not reflect requirements for MPUs. The exchanges accomplished in **Figure 2.2.2** are:

- **Skill Accepts Commands and Requests Resources** - Mission Management publishes capability commands which are received by the Sensor Skills.
  - When a Sensor Skill is commanded using a capability command, the Sensor Skill processes the command and publishes one or more activity messages associated with the command, plus status messages corresponding to the command. These messages acknowledge the receipt and status of the command.
  - The Sensor Skill then requests MFA resources for the activity using `RF_ResourceAllocationRequest`(s) messages. Each `RF_ResourceAllocationRequest` contains one or more candidate options for specific sets of MFA resources that the Sensor Skill could use for activity execution.
- **SRM Allocates Resources** - The SRM Resource Allocator evaluates the candidate options in the context of various factors, which include other existing allocations, current priorities, and mission state. Based on the evaluation, the Resource Allocator identifies a feasible option consistent with mission priorities and publishes an `RF_ResourceAllocation` message communicating the selected option.
- **Skill Initiates Command Execution** - After receiving the `RF_ResourceAllocation`, and an `RF_ResourceAllocationStatus` with a state of AVAILABLE, the Skill begins low latency interactions with the MFA using the MEL. The Skill continues reporting activity status using the activity messages over the ASB.
- **Skill Commands MFA** - MEL interchanges commence. The MEL interactions from the Sensor Skill flow directly to the MFA. Detailed MEL interactions are not captured in **Figure 2.2.2** and **Figure 2.2.3**. For more information on the RF MEL, see the RF MEL Interface Description Document (IDD).
- **MFA Responds to Commands and Provides Data** - Generally, the Sensor Skill publishes processed data products for consumption by other services. However, not all Sensor Skills will generate data.
- **Skill Terminates Command** - Activity functions continue until the Sensor Skill is commanded to stop or the activity has completed. Once an activity has ended, the resource allocations are removed and the activity state is transitioned to completed, deleted, or failed.

2.2.2.2 Response to New Task Request

The AMS GRA SRM framework is defined to enable dynamic resource allocations to multiple Sensor Skills that are contending for shared MFA resources. Following this approach, resources allocations may be modified, updated, or removed after they have been granted—even when in use. **Figure 2.2.3** presents an example information exchange that highlights the role of the SRM Skills associated with granting, modifying, and removing resource allocations based on changing mission conditions. In this use case, changes to allocations are in response to a new, higher priority request from a Sensor Skill, which drives a modification for a lower priority Skill's allocation. The change in allocation for the lower priority Skill is achieved through updated requests from the Skill. Alternatively, the lower

priority service may keep its allocation, but fail to achieve the scheduled actions due to priority rather than be re-allocated by the SRM. Neither pattern is prescriptive, and the behavior is left to MSI implementation, but these are intended to demonstrate the envisioned flexibility of the SRM framework.

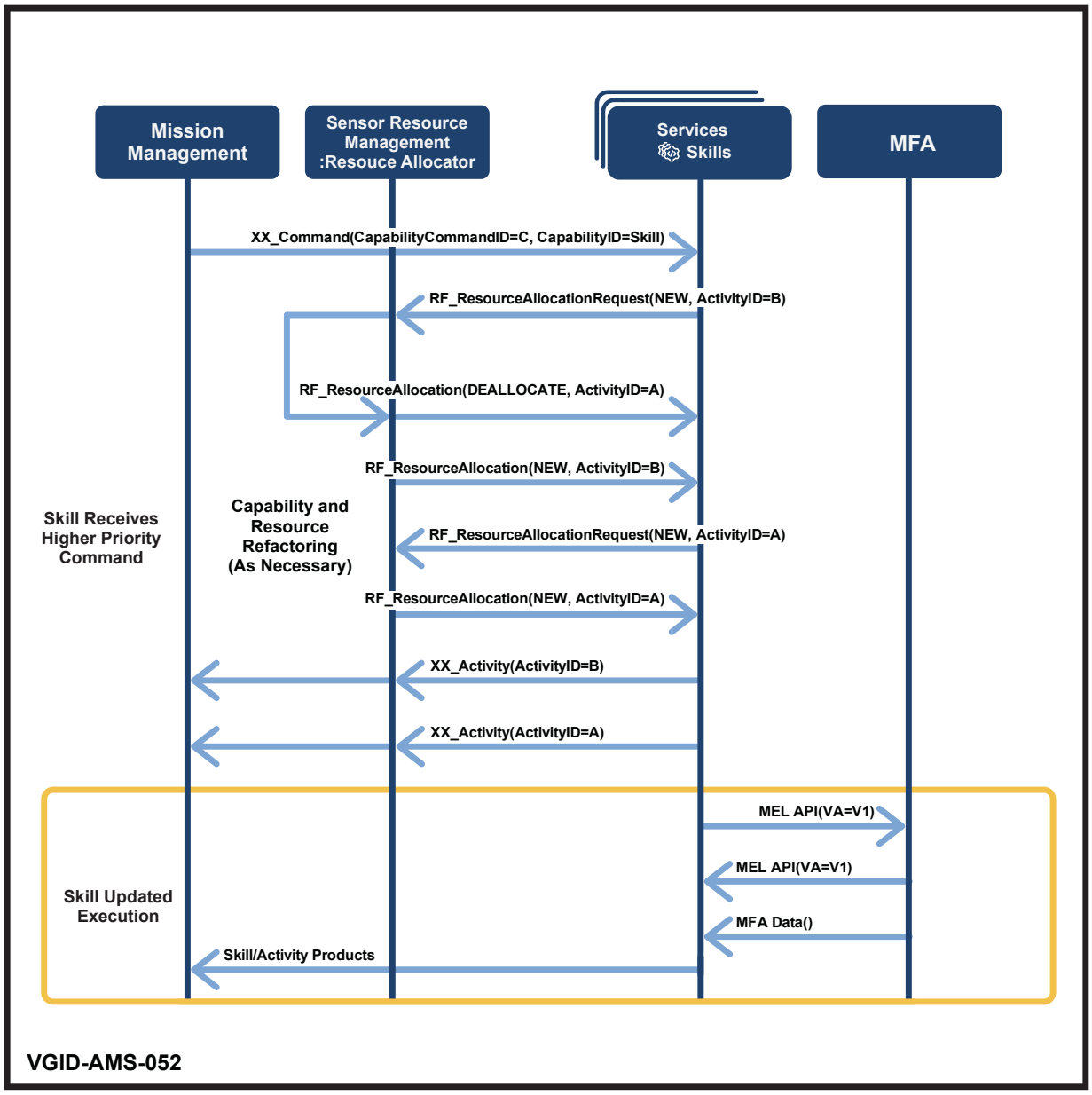


Figure 2.2.3: This SRM Sequence Diagram Shows How SRM Updates Resource Allocation Upon Receipt of Dynamic Mission Tasking.

This flow presents a relatively straightforward method for updating allocations, but more complex capabilities within the SRM Skills are also supported by the framework.

The exchanges in **Figure 2.2.3** are:

- **XX_Command** - Mission Manager generates a command and forwards it to the SRM and appropriate Skill(s).
- **RF_ResourceAllocationRequest** - Skill requests aperture resources from the SRM to fulfill a command.
- **Capability and Resource Refactoring** - SRM assigns new resource allocations to execute one or more tasks assigned to that MFA and possibly other interoperating COSITE MFAs.
- **RF_ResourceAllocation** - SRM forwards updated resource allocations to Skills. Skills evaluate performance impact of allocations.
- **Allocation Update** - If Skill determines that allocated resources are insufficient to meet minimal expected performance, Skill releases allocation via `RF_ResourceAllocationRequest(UPDATE)` message.
- **Job Request(s) (a.k.a. MEL Messaging)** - Each Skill schedules on-aperture time request.
- **MFA Data** - MFA streams Collected Data to requesting Skills for processing.
- **Skill/Activity Products** - Skills process Collected Data into products and disseminate to Resource Management and MPS.

2.2.2.3 Mission State Adaptation

An additional SRM behavior is to interact with Mission Management and Skills to update or change resource allocations to ongoing tasks in response to changes in the mission state. Examples of mission state changes include transitioning mission phases (which may change task priorities), and navigation or platform heading changes that affect the field-of-regard of an MFA such that it no longer can support an ongoing task. As these mission changes occur, Resource Management adapts resource allocations to sustain ongoing tasks with graceful transition. This adaptation is executed without requiring platform attention or intervention.

Processing and information exchanges among the actors to accomplish background adaptation includes:

- Mission Management tracks mission progression and conveys priority updates and significant changes of platform navigation or orientation state to SRM.
- SRM assesses impact of updates on priorities and resource allocations to active Skills, revises resource allocations as appropriate, and conveys revised allocations to the Skills.
- Skills adapt to revised resource allocations or propose alternative resource allocations.

Again, this adaptation is expected to occur on a relatively slow time scale and without pilot attention or intervention.

2.2.3 Data Model

The messaging listed in **Table 2.2.2** is extracted from the AMS GRA Supplemental Information Model and represents the key messaging identified in **Section 2.2.2**, Behavioral Architecture. For the complete message list and interface associations, please refer to the model. Note that this data model presently serves as an example and does not reflect requirements for MPUs.

Name	Documentation
MEL_Message	The information exchange to control and request MFA status for VA jobs.
MEL_Responses	The information exchange to respond to control and status requests from MFAs to Skills and SRM.
MFA_Data	Sensor information products to/from the MFA in the form of I/Q, PDWs, or other data obtained from or destined for sensors.
RF_Report	Indicates information about the past, current, and/or future RF usage of a capability.
RF_ResourceAllocation	Used for successful Resource Allocation. This message is issued from SRM to indicate the allowed resources granted to the Skill to perform its activities.
RF_ResourceAllocationRequest	Message type used with shared RF antenna subsystems to request a resource allocation from a Resource Manager.
RF_ResourceAllocationRequestStatus	This message indicates the result of an `RF_ResourceAllocationRequest`. This message will reflect whether the command was received, accepted, and/or rejected.
Skill/Activity Products	Post-processed information products from Skills to the Mission System for reasoning and communication.
XX_Activity	This message indicates the combined in-work and to-do list of the capability of a subsystem of a system.
XX_Command	Capability Command contains configuration parameters, tasking constraints, performance constraints, targets, and mission priority. This is the primary input to the SRM to command Skills to perform activities and request resources.

XX_CommandStatus	This message indicates the result of a command. This message will reflect whether the command was received, accepted, and/or rejected.

Table 2.2.2: These Key SRM Messages Reflect All Messages Referenced in the SRM Behavioral Architecture Sequence Diagrams.

2.3 Deployment Resource Management

Deployment Resource Management comprises initializing Skills, provisioning these Skills with necessary processing and memory allocations, establishing the data paths required to operate, and evaluating the overall Mission System processing capacity versus information needs. Deployment Management is implemented by Mission Agnostic Service Infrastructure (MASI). Refer to the Mission Agnostic Service Infrastructure Volume for more details.

A key aspect of MFP layer Deployment Resource Management is managing the processing and Avionics Backplane Bus (ABB) network resources used to host services and transfer high-bandwidth, low-latency data between the MFP services and MFAs, and among MFP services. This management includes:

- Selecting DPPs to host Services and their constituent Digital Payloads (DPs) IAW the deployment and recovery strategy, and budgeting of Central Processing Unit (CPU), memory, and network bandwidth resources on each DPP for the service
- Sizing allocated memory pools
- Ensuring that sufficient bandwidth is available across the ABB for the data transfer
- Monitoring and providing network and compute resource loading statistics

Although MFP layer Deployment Resource Management elements make decisions about how (i.e., what and when) to allocate processing and network related resources, the resource management elements do not necessarily implement the decisions. For example, if an MFP layer Resource Management element decides to deploy a service to three DPPs, it would tell another piece of infrastructure software to do the actual deployment. This separation of responsibility keeps the Resource Management logic isolated from the implementation details of the infrastructure and the diversity of deployment paradigms and mechanisms it may support, while keeping prioritization and decision logic out of the infrastructure.

2.4 General Resource Management

General Resource Management addresses multi-ship and own-ship Resource Management. For example, communications capacity management, multi-ship content routing, or coordination of multi-ship sensing. Considerations of negotiating conflicting resources between security and mission critical needs may also fall into this category when within the jurisdiction of the Mission System. General Resource Management works in conjunction with Deployment and SRM to coordinate and manage resource allocation and execution across a distributed team.

A.0 Acronyms

ABB	Avionics Backplane Bus
AMS	Agile Mission Suite
API	Application Programming Interface
ASB	Abstract Service Bus
C2	Command and Control
CPU	Central Processing Unit
DP	Digital Payload
DPP	Digital Payload Processor
GRA	Government Reference Architecture
IAW	In Accordance With
IDD	Interface Description Document
IFF	Identify Friend or Foe
IMPL	Implementation
IR	Infrared
JTRS	Joint Tactical Radio System
MASI	Mission Agnostic Service Infrastructure
MEL	MFA Encapsulation Layer
MFA	Multi-Function Aperture
MFP	Multi-Function Processor
MIDS	Multi-Functional Information Distribution System
MPS	Mission Processing Subsystem
MPU	Minimum Procurable Unit
MSI	Mission System Integrator
OEM	Original Equipment Manufacturer
OMS	Open Mission Systems
PDW	Pulse Descriptor Word
RF	Radio Frequency

SK	Starter Kit
SRM	Sensor Resource Management
Tx	Transmit
UCI	Universal Command and Control Interface
VA	Virtual Aperture

B.0 Revision History

The Revision History table, **Table B.0.1**, captures the major changes made for a given AMS GRA volume as the Starter Kit (SK) continues to mature.

Revision	Description	Date	Notes
-	This revision adds a revision table to the volume.	May 22, 2024	SK11.0
A	This revision adds: 1. Authoritative References table 2. Enhanced Figure and Table captions, 3. This revision removes the Resource Management BDD diagram.	July 24, 2025	SK13.0
B	This revision did not introduce any significant changes.	April 24, 2026	AMS GRA 14.0

Table B.0.1: Revision History Captures the Major Changes Made to the Resource Management Volume Between Documentation Releases.